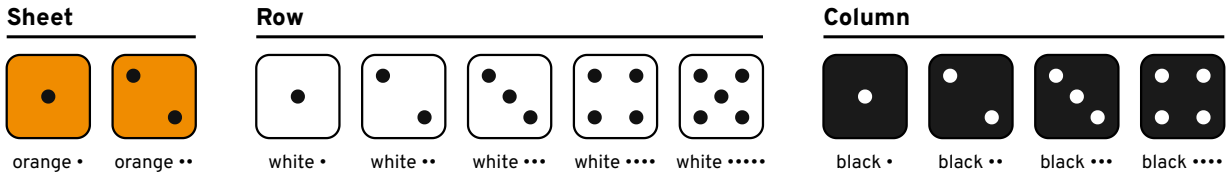
KEY dice ·the eleven dice

Eleven six-sided dice, each of which works as a coin. One throw of all eleven gives eleven yes-or-no answers, which is one row of the seed plate. Every bit of the key's randomness comes from these dice.



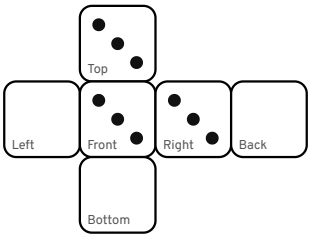
Each die is a coin

Three faces carry dots and three are blank. Every dotted face is opposite a blank one, so a throw is exactly 50/50.

Dots on top = ● = 1 = mark the cell.
Blank on top = ○ = 0 = leave it empty.

Read the top face only. How many dots it shows does not matter for the result.

One die, unfolded



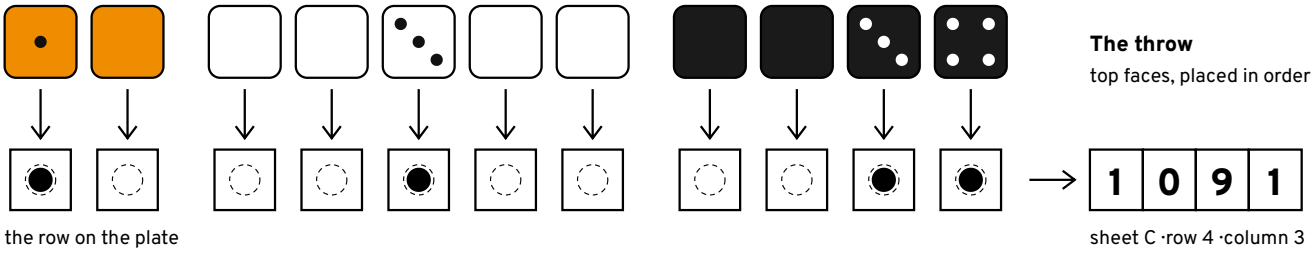
The white three-dot die. All three marked faces show the same three dots. They meet at one corner, so two of them are always visible from the side, whichever face lands on top.

Each die has one place

The dots are a name, not a score. The **colour** says which group a die belongs to, and the **number of dots** says its position in that group, fewest on the left. Nobody chooses where a die goes, so a result cannot be steered.

Group	Dice	Dots	On the number sheets	On the plate row
Sheet	2 orange	1, 2	which sheet, A to D	cells 1 and 2
Row	5 white	1 to 5	which row, 0 to 31	cells 3 to 7
Column	4 black	1 to 4	which column, 0 to 15	cells 8 to 11

One throw, one row



Eleven coins give 2,048 possible rows, all equally likely, numbered 0000 to 2047. A key takes **twelve throws**: eleven of all the dice, then one of the two orange and five white dice only. The last four cells of row 12 come from LOCK.

How they are made

- 16 mm, opaque, rounded corners.
- Dots are printed flush, never drilled. Drilled dots would make those faces lighter and tilt every throw.
- No numbers, logos or other marks.
- Black dots on orange and white, white dots on black. The three colours differ in brightness as well as hue, so they can be told apart in poor light or with colour blindness.

With ordinary dice

Any six-sided die works as a coin: **even = ●, odd = ○.**

Ordinary dice do not name their own place, so the order must be fixed another way: throw them one at a time and fill the strip from left to right.

Used with: the dice strips, which hold a throw in order; the number sheets A to D; the seed plate; and the run sheet, which gives the steps.